



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
(MECHANICAL ENGINEERING)
EMM 305: METROLOGY

3RD TRIMESTER 2022/2023

TIME: 2 HOURS

By Mr. Sammy Retich

INSTRUCTIONS:

- i) This paper contains **FIVE (5)** questions.
- ii) You are required to answer **THREE (3)** questions only.
- iii) Question **ONE** is compulsory.
- iv) Attempt any **OTHER TWO** question.

QUESTION ONE (30 MARKS) COMPULSORY

- a) Using suitable sketches define the following terms. (5 Marks)
 - i) Accuracy
 - ii) Precision
- b) Outline **FOUR** accuracy requirements of a sine bar.
- c) Define and state **THREE** applications of angle dekkor. (5 Mark)
- d) A slip gauge is being inspected using the NPL flatness interferometer. It is recorded that the gauge exhibits 10 fringes along its width in one position and 18 fringes in the other position. If the wavelength of the monochromatic light source is $0.5 \mu\text{m}$, determine the error of flatness over its width. (4 Marks)
- e) Describe the measurement of lead in a spur gear (5 Marks)
- f) Discuss the various metrological applications of an optical projector (4 Marks)
- g) Explain how the straightness of a machine guideway is assessed using an Autocollimator. (7 Marks)

QUESTION TWO (20 Marks)

- a) With the aid of sketches where necessary describe a laboratory method of carrying out the measurements using a sine bar measuring unknown angles of heavier components with more accuracy. (7 Marks)
- b) State FOUR ^{limitations using sine bar} ~~disadvantages of materials.~~ (4 Marks)
- c) With the aid of sketches, describe the working principle of a Zeiss Ultra-optimeter mechanical comparator. (9 Marks)

QUESTION THREE (20 Marks)

- a) With the use of suitable sketches, define the following terms as used in metrology (5 Marks)
- i) Roughness
 - ii) Waviness
 - iii) Lay
- b) It is practically impossible to produce a component that is free from surface irregularities. What could be the cause of these imperfections of a surface which are in the form of succession of hills and valleys varying in both height and spacing? (5 Marks)
- c) With the aid of sketches where necessary, explain the working principle of flatness testing interferometer. (10 Marks)

QUESTION FOUR (20 MARKS)

- a) State FIVE applications of laser interferometry. (5 Marks)
- b) With the aid of sketches describe the following types of resistance strain gauges (9 Marks)
- i) Wire-type strain gauge
 - ii) Foil-type strain gauge
 - iii) Semiconductor or piezoresistive strain gauge
- c) Explain how the strain gauge is influenced by temperature variations. (3 Marks)
- d) State THREE roles of quality assurance department in a production/manufacturing enterprise. (3 Marks)

QUESTION FIVE (20 MARKS)

- a) Using suitable sketches, differentiate between active and passive transducers (8 Marks)
- b) With the aid of sketch, describe how pitch error is measured. (6 Marks)
- c) A tooth span micrometer is being used to measure the span across three teeth. Three trials are conducted and the average value of span width is 31.120 mm. The following data is available for the gear being inspected: the number of teeth = 32, addendum circle diameter = 136mm, pressure angle = 20° , and span of measurement = 3. Determine the percentage error of measurement. (6 Marks)